

ARTI 404 – Image Processing

Lab 3 Report

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1. Introduction

In this lab, OpenCV was used to apply different image transformations. The same input image was used for all the transformations.

2. Geometric Transformations

First, the size of the image was increased by two times using the resize function.

Then, the image was rotated by 120 degrees using a rotation matrix.

After that, a shear transformation was applied to change the shape of the image horizontally.

3. Intensity Transformations

A negative image was created by subtracting the image pixels from 255.

The log transformation was also applied to change the intensity values of the image. A suitable constant was calculated based on the maximum pixel value.

Finally, the power law transformation was applied using $\gamma = 0.5$. This increased the brightness and contrast of some areas of the image.

4. Results

The transformations produced different versions of the original image. The geometric transformations changed the size, direction, and shape of the image, while the intensity transformations changed the pixel intensity values.

5. Conclusion

In this lab, I learned how to use OpenCV to perform basic geometric and intensity transformations. These operations are useful for changing and improving images during image processing.